Is Alice Corp. v. CLS Bank Good for Innovation in the Software Industry? Evaluating USPTO and Congressional Responses

Introduction

The Supreme Court's unanimous decision in Alice Corp. v. CLS Bank International fundamentally changed the landscape of software patent law in the United States.[1] The case addressed whether computer-implemented inventions, particularly those involving abstract ideas, qualify for patent protection under 35 U.S.C. § 101. The Court held that Alice Corporation's patents for a computerized trading platform were invalid because they claimed abstract ideas without sufficient additional elements to transform them into patent-eligible inventions.[2] This decision has sparked intense debate about its impact on innovation in the software industry. While proponents argue that Alice eliminates low-quality patents that stifle innovation, critics contend that it creates uncertainty and discourages investment in software development. The decision's effects ripple through various stakeholders, from individual inventors to major technology companies, raising fundamental questions about the balance between protecting intellectual property and fostering technological progress.

The Alice decision represents more than just a ruling on specific patents; it embodies a fundamental shift in how the legal system approaches the intersection of technology and intellectual property law. The case emerged from decades of confusion about software patent eligibility, which had created a fractured legal landscape where similar inventions might receive vastly different treatment depending on how they were presented to patent examiners or courts. This uncertainty had real economic consequences, as companies struggled to make informed decisions about research and development investments without clear guidance on what types of innovations would receive patent protection. The Supreme Court's intervention was both

[1] Alice Corp. v. CLS Bank Int'l, 134 S. Ct. 2347, 2352 (2014).
[2] Id. at 2359-60.

necessary and controversial, as it attempted to bring clarity to an area of law that had become increasingly complex and unpredictable.

Historical Context and Evolution of Software Patent Law

Moving from rejecting software patents entirely, the law started to widely accept them before returning to a restrictive framework under what is now known as Alice. Early Supreme Court cases like Gottschalk v. Benson and Parker v. Flook established that abstract mathematical algorithms could not be patented simply by implementing them on computers.[3] However, Diamond v. Diehr opened the door for software patents by allowing protection for processes that used mathematical formulas as part of larger industrial applications.[4] The Federal Circuit's decision in State Street Bank v. Signature Financial Group dramatically expanded software patent eligibility by establishing the "useful, concrete, and tangible result" test, which allowed patents for business methods implemented on computers.[5] This expansion led to an explosion in software patent applications and grants throughout the 1990s and 2000s.

After State Street, the number of software patents being filed increased greatly and the USPTO granted many thousands of patents that covered both simple online shopping and basic computer systems. As a result, experts began talking about a "patent thicket," because firms had to study many existing patents before they could come up with new software solutions.[6] Matters became much worse in industries where technology advanced swiftly, because innovators started with others' ideas and patent claims generally targeted wide ideas instead of tiny improvements. The emergence of patent assertion entities, commonly called patent trolls, further complicated the

[3] Gottschalk v. Benson, 409 U.S. 63, 68 (1972); Parker v. Flook, 437 U.S. 584, 594-95 (1978).
[4] Diamond v. Diehr, 450 U.S. 175, 192-93 (1981).
[5] State St. Bank & Trust Co. v. Signature Fin. Grp., Inc., 149 F.3d 1368, 1375 (Fed. Cir. 1998).
[6] Brian J. Love et al., Brief of Amici Curiae Law, Business, and Economics Scholars in Alice Corp. v. CLS Bank, at 26-27 (2014).

landscape by creating a secondary market for software patents that emphasized litigation potential over technological contribution.

The Supreme Court's Decision and Reasoning

The Supreme Court unanimously affirmed the Federal Circuit's judgment, establishing a two-step framework for analyzing patent eligibility under § 101.[7] First, courts must determine whether the claims are directed to patent-ineligible concepts such as laws of nature, natural phenomena, or abstract ideas. Second, if the claims involve such concepts, courts must examine whether the claims contain additional elements that transform the abstract idea into a patent-eligible application.[8] The Court found that Alice's patents were directed to the abstract idea of intermediated settlement, which had been a fundamental economic practice long prevalent in commerce.[9] The Court rejected the argument that implementing this abstract idea on a computer automatically made it patent-eligible, stating that "the mere recitation of a generic computer cannot transform a patent-ineligible abstract idea into a patent-eligible invention."[10] This decision built upon previous Supreme Court precedents in Mayo Collaborative Services v. Prometheus Laboratories and Bilski v. Kappos, which had similarly restricted the scope of patentable subject matter.

The Court's reasoning reflected deep concerns about the monopolization of fundamental concepts that should remain available for public use. Justice Thomas's majority opinion emphasized that patent law must strike a careful balance between incentivizing innovation through exclusive rights and preserving access to the basic tools of scientific and technological

[7] Alice Corp., 134 S. Ct. at 2355.
[8] Id.
[9] Id. at 2356.
[10] Id. at 2358.

work.[11] The decision drew heavily on the preemption doctrine, which holds that patents should not foreclose entire fields of inquiry or prevent others from using fundamental principles.[12] The Court's analysis also reflected skepticism about the proliferation of patents that claimed traditional business methods merely by adding computer implementation, suggesting that such patents often failed to advance the useful arts in meaningful ways.

Detailed Analysis of Arguments Supporting Alice

Those backing the Alice verdict state that it has helped the system become more effective and decreased challenges for innovators. Based on their research, Bessen and Meurer observe that software patents usually resulted in less investment in research and development rather than more innovation.[13] From their investigations, it appears that companies used patents mostly to protect their positions, while very little resource was dedicated to producing actual inventions. Thanks to Alice, there are fewer abstract patents granted which reduces why some businesses must accumulate unneeded patents to stay safe from lawsuits.

Now, with the decision, small software companies and startups have seen their challenges from overly broad patents reduced significantly. Mobile and web entrepreneurs, as well as software developers, were previously stuck by basic patents related to things like payment processing, verifying a user’s identity or synchronizing data.[14] Often, the broad concepts claimed in these patents prevented new companies from bypassing them and gave rise to questions about just how valid they were. Many software patents have been declared invalid by the Alice framework, opening up the way for more innovation in basic software practices.

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[11] Id. at 2354.
[12] Id. at 2354-55.
[13] Love et al., supra note 6, at 8-11.
[14] Id. at 14-17.

Thanks to Alice, patent examiners now have to check if an application describes an abstract idea and if further elements give enough limitations.[15] The extra carefulness has inspired examiners to check more closely how software patents contribute to technology. The change has caused the number of patents that merely cover ordinary computer activities or basic business methods to decrease. Those watching the industry think that patents given after Alice center on resolving technical issues rather than general business ideas.

With this decision, it has become more challenging for patent assertion firms to grab money from real companies using legal threats. Recent research shows that there has been a big decrease in PAE suits since Alice, mainly in cases about software patents for abstract business methods.[16] Because nuisance lawsuits are less common now, firms can use their time and money on new ideas instead of constantly fighting over patents. Because of the change, weak patents are easier to spot by courts at the beginning of litigation, so litigation costs and uncertainty are reduced.

Detailed Analysis of Arguments Against Alice

Some argue that the decision will discourage both innovation and new investments in creating software. With the decision’s general terms, people are unsure about which software inventions can receive a patent.[17] Venture capital funding faces a major problem because investors are required to examine the intellectual property of startup companies. Many people in the venture capital field say it is harder to assess software companies because they don’t know if their latest creations will be able to get a patent.[18]

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[15] Jordana Goodman, Alice Corp. v. CLS Bank Int'l, 21 B.U. J. Sci. & Tech. L. 224, 230-31 (2015).
[16] Love et al., supra note 6, at 16-17.
[17] John Clizer, Exploring the Abstract: Patent Eligibility Post Alice Corp v. CLS Bank, 80 Mo. L. Rev. 537, 551-52 (2015).
[18] Charles F. Green, CLS Bank v. Alice Corp.: What Does It Mean For Software Patent Eligibility?, 13 J. Marshall Rev. Intell. Prop. L. 601, 620 (2014).

The pharmaceutical and biotechnology industries are especially concerned that Alice might be applied to their areas. Under Alice, computer analysis of biological data has become more closely examined and some judges have ruled that, in these cases, patent protection should not be applied.[19] Such a trend might stop major investments in personalized medicine and computer-aided drug discovery, because they rely on patents for financial justification. Alice's abstract idea test is so widely written that companies in various technology industries cannot determine the clear limits of patent eligibility.

Alice critics are also worried about China's ability to compete on the global stage. In the United States, only a limited number of software inventions can be patented, whereas other main patent jurisdictions easily grant these sorts of patents.[20] Because the European Patent Office recognizes that software innovation can also deal with technical issues, it allows patents for applications in business or abstract areas. Japan and South Korea have laws that permit the patenting of many software inventions. The difference between the United States and other countries could make it harder for American companies to safeguard their inventions in the global market.

The ruling could lead to more trade secret protection than sharing patents which could slow the progress made by patents for the public. When a business does not have the option to patent new software, they may decide to keep the information private.[21] If patent disclosure changes, it could slow innovation by limited knowledge being shared with others. The disclosure purpose of patents benefits the public by allowing researchers to access technical information needed for invention and development.

USPTO's Comprehensive Response Strategy

[19] Clizer, supra note 17, at 555-56.
[20] Green, supra note 18, at 615.
[21] Clizer, supra note 17, at 554.

The United States Patent and Trademark Office responded to Alice with an extensive series of guidance documents and procedural changes aimed at providing consistent application of the new eligibility standards. The initial guidance issued one week after the Supreme Court decision provided emergency instructions for handling pending applications, but the USPTO quickly recognized the need for more comprehensive guidance.[22] The agency's 2014 Interim Guidance on Patent Subject Matter Eligibility represented the first systematic attempt to translate Alice's framework into practical examination procedures.[23]

The USPTO's guidance documents attempt to provide concrete examples of patent-eligible and patent-ineligible claims across various technology areas. These examples include software inventions that improve computer functionality, such as enhanced data compression algorithms or improved user interface designs, which generally qualify for patent protection under Alice.[24] Conversely, the guidance identifies business method patents that merely automate traditional processes without technological improvements as likely patent-ineligible. The agency has continued to refine these examples through multiple updates, reflecting ongoing efforts to achieve consistency in examination practices.

Training programs for patent examiners have been substantially expanded to address subject matter eligibility issues. The USPTO has developed specialized curricula focused on identifying abstract ideas and evaluating additional elements that might transform abstract concepts into patent-eligible applications.[25] These training programs include detailed case studies and hypothetical scenarios designed to help examiners apply Alice consistently across different

[22] Memorandum from Andrew H. Hirshfield, Deputy Commissioner on Patent Examination Policy on Examination Instructions in View of the Supreme Court Decision in Alice Corp. Pty. Ltd. v. CLS Bank Int'l (Jun. 25, 2014).
[23] 2014 Interim Guidance on Patent Subject Matter Eligibility, 79 Fed. Reg. 74,618 (Dec. 16, 2014).
[24] Id.
[25] Goodman, supra note 15, at 231.

technology areas. The agency has also established quality review procedures specifically focused on § 101 determinations to ensure uniform application of eligibility standards.

The Patent Trial and Appeal Board has played a crucial role in developing precedent for Alice's application. PTAB decisions have provided guidance on specific technology areas and claim types, helping to clarify the boundaries of patent eligibility.[26] These decisions have addressed issues such as the treatment of user interface patents, database management systems, and artificial intelligence applications. The PTAB's precedential decisions have helped establish more predictable standards for patent eligibility determinations.

Empirical Evidence of Prosecution and Examination Changes

Analysis of statistics shows that the way patent applications are handled has changed significantly since Alice was decided. According to Robert Sachs' research, acceptance of business method patents dropped dramatically from roughly 10% in pre-Alice times to only 10% in some technology centers.[27] Software patents, in general, are being rejected at a rate of 70-80% compared to the previous overall rate of under 20%. This data shows that Alice has made a strong impact on patent examinations and so far has led to many inventions being challenged on their eligibility for patents.

As a result of increasing rejection rates, it now takes more time and costs more money for people to get their patents. Previously, competitors would have been checked mainly against novelty, but today, they often face a first § 101 rejection that may take many attempts to overcome.[28] Patent attorneys report spending significantly more time crafting initial applications to address potential

[26] Id.
[27] James E. Daily, Alice's Aftermath: Changes in Patentee Behavior Since Alice v. CLS Bank, 23 B.U. J. Sci. & Tech. L. 284, 286 (2017).
[28] Id.

eligibility issues as well as to responding to § 101 rejections during prosecution. These increased costs may end up disproportionately affecting small inventors as well as companies with limited resources for patent prosecution.

Abandonment rates for software patent applications have increased substantially since Alice, with many applicants choosing to discontinue prosecution rather than pursue patents in uncertain legal territory.[29] This trend is particularly pronounced for business method applications, where applicants face rejection rates exceeding 90% in some technology areas. The high abandonment rates suggest that many inventors have concluded that the costs and uncertainties of patent prosecution outweigh the potential benefits of patent protection.

Appeal rates to the Patent Trial and Appeal Board have also increased significantly for § 101 rejections, indicating that many applicants dispute examiner determinations of patent ineligibility.[30] However, success rates on appeal remain relatively low for applications rejected under Alice, suggesting that the PTAB generally supports examiner determinations of patent ineligibility. These appeal statistics provide additional evidence of the substantial impact Alice has had on patent prosecution practices.

Congressional Responses and Legislative Initiatives

Congressional response to Alice has been characterized by extensive debate but limited legislative action. Multiple bills have been introduced to modify § 101 or provide clearer standards for patent eligibility, reflecting widespread concern about the decision's impact on innovation.[31] The STRONGER Patents Act, introduced in several congressional sessions, would

[29] Id. at 294-95.
[30] Id. at 286.
[31] Green, supra note 18, at 620.

eliminate the current judicial exceptions for abstract ideas, laws of nature, and natural phenomena, instead relying solely on novelty, non-obviousness, and disclosure requirements to screen patents.[32] Supporters argue that this approach would provide greater certainty for inventors while maintaining adequate quality controls through existing patentability requirements.

The Intellectual Property Owners Association has developed detailed legislative proposals that would specifically overturn Alice while maintaining restrictions on the most problematic patents.[33] Their proposed amendments to § 101 would establish that inventions are patent-eligible if they belong to one of the four statutory categories, regardless of whether they involve abstract ideas or other judicial exceptions. However, the proposals would maintain restrictions on patents that claim fundamental principles or natural phenomena without practical applications.

Industry lobbying efforts have reflected deep divisions about the appropriate scope of patent protection. Technology companies like Google, Microsoft, and Amazon have generally supported Alice's restrictions on software patents, arguing that broad patent eligibility stifles innovation and increases litigation costs.[34] These companies contend that software innovation thrives through rapid iteration and improvement rather than patent protection, and that overly broad patents create barriers to technological progress. Conversely, pharmaceutical and biotechnology companies have advocated for broader patent eligibility, expressing concern that Alice's reasoning could extend to invalidate patents for computer-aided drug discovery and diagnostic methods.[35]

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[32] Nathan Peske, CLS Bank International v. Alice Corp. Pty. at the Federal Circuit: The Dilemma Presented by Computer Implementation of Abstract Ideas and How the Supreme Court Missed a Chance to Clear It Up, 16 Minn. J.L. Sci. & Tech. 509, 544-45 (2015).
[33] Clizer, supra note 17, at 556.
[34] Love et al., supra note 6, at 41-53.
[35] Clizer, supra note 17, at 555-56.

The biotechnology industry's concerns about Alice reflect the significant role that computer analysis plays in modern biological research. Many diagnostic methods rely on computer algorithms to analyze biological data and identify disease markers or predict treatment responses.[36] Patents covering such methods have faced increased scrutiny under Alice, with some courts finding them patent-ineligible as abstract mental processes. The industry argues that restricting patent protection for such innovations could discourage investment in personalized medicine and precision therapeutics.

Sectoral Analysis of Innovation Impact

Different sectors of the software industry have experienced varying impacts from Alice, revealing the decision's uneven effects across technology areas. Financial technology companies have faced particularly severe challenges, as many fintech innovations involve business methods implemented through software.[37] Patents covering payment processing, risk assessment, and financial transaction management have frequently been invalidated under Alice, forcing fintech companies to rely more heavily on trade secrets and first-mover advantages. This shift has accelerated innovation cycles in financial technology, as companies compete through rapid product development rather than patent protection.

Gaming and entertainment software companies have generally adapted more successfully to Alice's requirements. Patents covering specific game mechanics, user interface innovations, and rendering technologies often satisfy Alice's standards by focusing on technical improvements to computer functionality.[38] Companies like Electronic Arts and Activision Blizzard have maintained substantial patent portfolios by emphasizing the technical aspects of their innovations

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[36] Id.
[37] Goodman, supra note 15, at 231.
[38] Green, supra note 18, at 617-18.

rather than abstract game concepts. The gaming industry's experience suggests that software patents can survive Alice when they demonstrate clear technological contributions.

Artificial intelligence and machine learning present particularly interesting cases for Alice analysis. Patents covering specific algorithmic improvements or novel neural network architectures have generally survived Alice challenges when they focus on technical implementations rather than abstract mathematical concepts.[39] However, patents claiming broad approaches to artificial intelligence or generic machine learning methods have faced significant eligibility challenges. The AI industry has responded by focusing patent applications on specific technical implementations and demonstrable improvements to computer performance.

E-commerce and online business platforms have seen substantial changes in patent strategy since Alice. Companies like Amazon and eBay have shifted focus from broad business method patents to more specific technical innovations in areas like database optimization, user interface design, and system architecture.[40] The "one-click" purchasing patent, which would likely face eligibility challenges under Alice, represents the type of broad business method patent that the decision was designed to eliminate. The e-commerce industry's adaptation demonstrates how companies can maintain intellectual property protection by focusing on technical rather than business method innovations.

Mobile application development has been significantly affected by Alice, as many app-based innovations involve user interface designs or business methods that may not satisfy current eligibility standards.[41] App developers have increasingly relied on copyright protection for user interface elements and trade secret protection for algorithms rather than seeking patent

[39] Goodman, supra note 15, at 231-32.
[40] Love et al., supra note 6, at 20-21.
[41] Green, supra note 18, at 619.

protection. This shift has contributed to faster innovation cycles in mobile software development, as companies compete through rapid feature development and user experience improvements rather than patent portfolios.

International Comparative Analysis and Competitive Implications

The divergence between United States patent law and international approaches to software patent eligibility has created significant competitive implications for American companies. The European Patent Office continues to grant patents for computer-implemented inventions under its established "technical contribution" standard, which examines whether inventions solve technical problems through technological means.[42] This approach allows patent protection for many software innovations that would face eligibility challenges under Alice, potentially giving European companies advantages in global patent portfolios.

Japan's approach to software patent eligibility emphasizes the technical nature of claimed inventions rather than categorical exclusions for abstract ideas.[43] Japanese patent law allows protection for business method innovations when they involve technical implementations, and the Japan Patent Office has not adopted Alice-style restrictions on software patents. This permissive approach has attracted increased patent filing from international companies seeking to protect software innovations that might not qualify for protection in the United States.

South Korea has similarly maintained broad patent eligibility for software innovations, with the Korean Intellectual Property Office granting patents for computer-implemented business methods and software applications.[44] The Korean approach focuses on novelty and inventive step

[42] Peske, supra note 32, at 512.
[43] Green, supra note 18, at 615.
[44] Id.

requirements rather than categorical subject matter exclusions, allowing patent protection for many types of software innovations. This policy has supported South Korea's growth as a major technology center and has attracted investment from multinational companies seeking comprehensive patent protection.

China's rapid development as a technology center has been accompanied by increasingly broad patent protection for software innovations. The China National Intellectual Property Administration has established guidelines that generally allow patent protection for computer-implemented inventions when they demonstrate technical effects or solve technical problems.[45] This approach has supported China's goals of encouraging domestic innovation and attracting foreign technology investment, while potentially disadvantaging American companies that cannot obtain equivalent patent protection in their home market.

The practical implications of these international differences extend beyond patent portfolio management to fundamental business strategy decisions. Multinational technology companies must now consider patent law variations when deciding where to locate research and development activities.[46] Some companies have shifted software development to countries with more favorable patent regimes, while others have adapted their innovation strategies to focus on areas that qualify for patent protection across multiple jurisdictions.

Economic Analysis and Innovation Metrics

Comprehensive economic analysis of Alice's impact reveals mixed effects on innovation metrics and investment patterns. While venture capital investment in software companies has increased

[45] Clizer, supra note 17, at 552.
[46] Green, supra note 18, at 620.

since Alice, this does not seem to have officially stopped innovation funding.[47] However, developers find it increasingly tough to assess software companies because of unclear IP protection. As a result, investors might choose to support companies that hold strong trade secrets or who were first to market, rather than those that focus mainly on patents.

Patterns of research and development spending in different areas of technology have varied widely since Alice. Firms in areas where Alice matters such as financial and business technology, have continued or increased research spending while working on new technologies more likely to receive patents.[48] Shifting investments toward technology-focused fields may really make innovation more efficient.

Revenue from licensing patents has dropped in some technology fields and this can show that either the company's patents are less valuable or they are less motivated to invent. At the same time, the effect on new innovations appears minimal since firms now find other ways to realize their value.[49] Among these strategies are producing products more quickly, better trade secret protection and prioritizing brand and experience differences for users.

The system for startups has kept moving forward despite Alice's patents, since there is ongoing growth in forming new software companies and giving early-stage support.[50] It seems that in rapidly changing markets, where early adopters and market share are the main concern, software inventors may feel patent protection is not so crucial. Software entrepreneurship continues to be healthy, showing that good ideas can still grow despite protection by patents.

USPTO Budget and Administrative Impact

[47] Daily, supra note 27, at 287.
[48] Love et al., supra note 6, at 8-11.
[49] Daily, supra note 27, at 300.
[50] Id.

Alice's impact on USPTO has affected many areas at the USPTO such as budget planning and the process of how resources are managed. Thanks to this decision, the agency must now spend more on training patent examiners about subject matter eligibility.[51] To maintain Alice's standards in different technology centers, these training programs depend on spending a lot on curriculum development, instructors' time and ongoing educational support.

The time taken to examine software patent applications has gone up, as it is now necessary for examiners to evaluate if subject matter is eligible along with usual novelty and non-obviousness concerns. [52] As a result of these policy changes, examining patents now requires more examiner time, leading to longer waits in the patent prosecution process. Because of the increased responsibility linked to § 101 matters, the USPTO has hired more examiners and modified how examinations are performed.

Because there have been so many appeals about subject matter eligibility, the Patent Trial and Appeal Board has added more administrative judges and support staff.[53] Since technical and legal difficulties are common in these cases, solving them takes dedication and expertise. Technology-related appeals are now heard by special panels in the PTAB which have designed faster approaches to handling typical ineligibility dilemmas.

Programs for ensuring quality have been improved to guarantee that Alice standards are applied the same way in every examination unit. The USPTO has set up new review steps for handling § 101 cases and has determined performance standards for examiners on eligible subject matter.[54]

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[51] Daily, supra note 27, at 301-02.
[52] Goodman, supra note 15, at 230.
[53] Daily, supra note 27, at 286.
[54] Id. at 301.

Ensuring quality of examinations in this way means more supervision and checking is needed to keep the exams on track.

Future Considerations and Emerging Technologies

The rapid development of artificial intelligence, blockchain, and quantum computing technologies presents new challenges for applying Alice's framework to emerging innovations. AI patent applications often involve mathematical algorithms and data processing methods that may trigger abstract idea concerns under Alice, even when they represent significant technological advances.[55] The USPTO has begun developing specific guidance for AI patent applications, focusing on technical implementations and improvements to computer functionality rather than abstract algorithmic concepts.

Blockchain and cryptocurrency technologies present similar challenges, as many blockchain innovations involve business methods for conducting transactions or managing digital assets.[56] Patents covering blockchain implementations must demonstrate technical improvements or solutions to specific technological problems to satisfy Alice's requirements. The cryptocurrency industry has adapted by focusing patent applications on specific technical aspects of blockchain implementation rather than abstract concepts of digital currency or transaction processing.

Quantum computing represents an emerging area where Alice's application remains uncertain. Quantum algorithms and quantum information processing methods may involve abstract mathematical concepts, but they also represent fundamental advances in computing technology.[57] The USPTO will need to develop specific guidance for quantum computing patents to ensure that

[55] Green, supra note 18, at 618-19.
[56] Clizer, supra note 17, at 556-57.
[57] Peske, supra note 32, at 543.

legitimate technological innovations receive appropriate protection while maintaining Alice's restrictions on abstract ideas.

Internet of Things (IoT) technologies often involve combinations of software, hardware, and business method innovations that may trigger Alice concerns. IoT patents must carefully balance claims covering software functionality with sufficient technical implementation details to satisfy patent eligibility requirements.[58] The IoT industry has responded by emphasizing hardware implementations and specific technical solutions to connectivity and data processing challenges.

Potential Legal and Policy Reforms

Several potential reforms could address the uncertainties and problems created by Alice while maintaining appropriate restrictions on low-quality patents. Legislative clarification of § 101 could provide more specific guidance about patent eligibility while preserving appropriate limitations on abstract idea patents.[59] Such legislation could establish clearer definitions of abstract ideas and provide safe harbors for certain types of technological innovations.

Adjustments within the USPTO can enhance how steady and easy to predict patent eligibility judgments are. Better training for examiners, detailed notes about § 101 requirements and improved quality assurance steps at the USPTO could make § 101 mean the same thing in several units.[60] There is potential for the USPTO to build specialized groups of examiners who specialize in assessing software patent eligibility.

Judicial reforms could provide greater clarity through more detailed Supreme Court guidance or Federal Circuit precedent development. The Supreme Court could grant certiorari in additional §

[58] Green, supra note 18, at 618.
[59] Peske, supra note 32, at 544-45.
[60] Daily, supra note 27, at 301-02.

101 cases to provide more specific guidance about Alice's application to different technology areas.[61] The Federal Circuit may also provide clearer guidelines for software innovations and deciding their eligibility.

Solutions from international cooperation might lessen the growing gap between how software patents are granted in the United States and other countries. Harmonization initiatives through organizations like the World Intellectual Property Organization could help reduce the competitive disadvantages created by inconsistent patent law across different jurisdictions.[62]

Critical Assessment of Long-term Implications

Alice affects much more than how patents are prosecuted and litigated right now. With this decision, the patent system is changing how it encourages discovery while ensuring public use of essential inventions.[63] Because of this rebalancing, innovation, where to invest and competitiveness in technology might all change significantly.

By choosing to emphasize technology in the innovation decision, companies could motivate individuals to focus on practical and well-defined approaches to innovation. Firms are expected to put greater effort into core research and development than on acquiring patents for general business techniques.[64] This shift could enhance the effectiveness of innovation by encouraging companies to use resources on genuine technology, not on gathering patents.

Still, because Alice's terms are not clear and are applied arbitrarily, innovative businesses could face challenges with future decisions. Rapidly progressing technologies require strong rules for

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[61] Clizer, supra note 17, at 557.
[62] Green, supra note 18, at 620.
[63] Love et al., supra note 6, at 33-34.
[64] Id. at 17-18.

intellectual property so that companies can decide to advance their research over several years.[65] Not know how some software will be judged in the Patent Act may lead some to avoid investing in these projects and areas where venture capitalists rely heavily on patents.

The international competitive implications of Alice may become more pronounced over time as other countries maintain more permissive approaches to software patent protection. American companies may face increasing disadvantages in global patent portfolios, potentially affecting their ability to compete in international markets and attract foreign investment.[66] This divergence could influence decisions about where to locate research and development activities and how to structure international business operations.

Conclusion

Alice Corc. v. CLS Bank shows that it is difficult to allow both protection of ideas and development in the software industry. Many concepts that prevented progress with abstract business method patents have now been removed from the system. Weak patents are now harder for patent assertion entities to use against true businesses and software developers no longer have to struggle as much against broad patent claims for simple computer functions and ordinary operations.

Alice tends to do more than remove suspicious patents; it also improves the quality of all patents and their examination. The more thorough examination by the USPTO has led to enhanced disclosure of patent claim details which points the focus toward real technological achievements and away from unclear ideas. Because of Alice, companies can now focus on development and research rather than waste time and money buying defensive software patents.

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[65] Green, supra note 18, at 620.
[66] Peske, supra note 32, at 545.

Even so, Alice has placed various obstacles in the way that need to be acknowledged when judging its impact on innovation. The unclear status of patent eligibility has made it hard for companies to organize long-term research and development, mainly in sectors that used to depend heavily on patents for getting investment money. With doubts about software intellectual property protection, venture capital funders have more difficulty judging these companies which may affect their decisions on investing in startups.

As other leading countries take a broader approach to software patenting, Alice is increasingly influential when compared internationally. Rising difficulties in international patent portfolios could make it harder for American companies to be competitive abroad and draw foreign investors. Because different countries have their own patent laws, it becomes a challenge for multinational technology companies and can decide where their research and development will take place.

By reaction to Alice, the USPTO showed what the decision has both challenged and brought to the system. The agency's detailed guidance and education attempts to ensure that patent reviews stay consistent, but the high rates of objections and appeals still prove there is much confusion. The agency will find new approaches to following the Alice rules while still ensuring valid technological innovations are protected.

Since Congress has not taken any steps to reform patent law, significant changes to § 101 won't happen quickly, so the patent system will change mainly through administrative decisions and court judgments. People continue to argue about how much we need patents because they clearly disagree on whether patents should mainly stimulate innovation or act as guards against a single group taking basic discoveries for their own. Even though stakeholders want easier-tounderstand rules in the technology realm, these differences in opinions keep comprehensive reform out of reach.

What we have learned from studies about Alice makes it difficult to reach simple answers. Even as a result of the decision, software innovation and spending in most technology fields have expanded. Such resilience makes it conceivable that software innovation does not require heavy reliance on patents today, mainly in technology-centered markets.

Its success will be defined by whether it encourages new technology and stops the widespread patent abuse software companies faced before the decision. At present, it seems that Alice has advanced substantially toward both objectives, yet the lasting results will be clear only after the technology industry has progressed further with the new patent environment. The growth of new technologies such as artificial intelligence, blockchain and quantum computing will show if Alice's system can hold both new advances and correct limits on concepts.

All the evidence seems to point to the conclusion that Alice corrects deficiencies, but does not solve all the issues, in the patent system's treatment of software. As a result, there are fewer problematic patents, thus better overall patent quality, yet new issues and difficulties for decision-making on innovation and investment remain. How Alice is implemented by the USPTO, courts and possibly laws will decide if there is an even balance between respecting new technologies and benefiting public progress. Patent law changes have not stopped the vitality of the software industry, so we expect changes in the ways ideas are protected and business rivals compete.

BIBLIOGRAPHY

CASES

Alice Corp. Pty. Ltd. v. CLS Bank Int'l, 134 S. Ct. 2347 (2014).

Bilski v. Kappos, 561 U.S. 593 (2010).

CLS Bank Int'l v. Alice Corp. Pty. Ltd., 717 F.3d 1269 (Fed. Cir. 2013) (*en banc*).

CLS Bank Int'l v. Alice Corp. Pty. Ltd., 768 F. Supp. 2d 221 (D.D.C. 2011).

CLS Bank Int'l v. Alice Corp. Pty. Ltd., 685 F.3d 1341 (Fed. Cir. 2012).

Content Extraction & Transmission LLC v. Wells Fargo Bank, 768 F.3d 1343 (Fed. Cir. 2014).

DDR Holdings, LLC v. Hotels.com, 773 F.3d 1245 (Fed. Cir. 2014).

Diamond v. Diehr, 450 U.S. 175 (1981).

Gottschalk v. Benson, 409 U.S. 63 (1972).

Intellectual Ventures I LLC v. Symantec Corp., 838 F.3d 1307 (Fed. Cir. 2016).

Mayo Collaborative Servs. v. Prometheus Labs., Inc., 132 S. Ct. 1289 (2012).

Parker v. Flook, 437 U.S. 584 (1978).

State St. Bank & Trust Co. v. Signature Fin. Grp., Inc., 149 F.3d 1368 (Fed. Cir. 1998).

STATUTES

35 U.S.C. § 101 (2012).

35 U.S.C. § 102 (2012).

35 U.S.C. § 103 (2012).

35 U.S.C. § 112 (2012).

GOVERNMENT PUBLICATIONS

Federal Regulations

2014 Interim Guidance on Patent Subject Matter Eligibility, 79 Fed. Reg. 74,618 (Dec. 16, 2014).

USPTO Memoranda and Guidelines

Memorandum from Andrew H. Hirshfield, Deputy Commissioner on Patent Examination Policy on Examination Instructions in View of the Supreme Court Decision in *Alice Corp. Pty. Ltd. v. CLS Bank Int'l* (Jun. 25, 2014).

Memorandum from Robert W. Bahr, Deputy Commissioner for Patent Examination Policy on Formulating a Subject Matter Eligibility Rejection and Evaluating Applicant's Response to a Subject Matter Eligibility Rejection (May 4, 2016).

July 2015 Update: Subject Matter Eligibility, U.S. Patent and Trademark Office (2015).

BOOKS AND TREATISES

James Bessen & Michael J. Meurer, *Patent Failure: How Judges, Bureaucrats, and Lawyers Put Innovators at Risk* (2008).

Robert W. Gomulkiewicz, *Software Law and its Application* (3d ed. 2023).

LAW REVIEW ARTICLES

John Clizer, Exploring the Abstract: Patent Eligibility Post *Alice Corp v. CLS Bank*, 80 Mo. L. Rev. 537 (2015).

James E. Daily, Alice's Aftermath: Changes in Patentee Behavior Since *Alice v. CLS Bank*, 23 B.U. J. Sci. & Tech. L. 284 (2017).

James E. Daily & F. Scott Kieff, Anything Under the Sun Made by Humans: Patent Law Doctrines as Endogenous Institutions for Commercializing Innovation, 62 Emory L.J. 967 (2013).

Sue Ann Ganske, *Alice Corp. v. CLS Bank Int'l*, 21 B.U. J. Sci. & Tech. L. 224 (2015).

Jordana Goodman, *Alice Corp. v. CLS Bank Int'l*, 21 B.U. J. Sci. & Tech. L. 224 (2015).

Charles F. Green, CLS Bank v. Alice Corp.: What Does It Mean For Software Patent Eligibility?, 13 J. Marshall Rev. Intell. Prop. L. 601 (2014).

Nathan Peske, CLS Bank International v. Alice Corp. Pty. at the Federal Circuit: The Dilemma Presented by Computer Implementation of Abstract Ideas and How the Supreme Court Missed a Chance to Clear It Up, 16 Minn. J.L. Sci. & Tech. 509 (2015).

Pamela Samuelson, The Case Against Copyright Protection for Computer Programs in Machine Readable Form, 1984 Duke L.J. 661 (1984).

AMICUS BRIEFS

Brief of Amici Curiae Electronic Frontier Foundation in Support of Grant of Petition, *CLS Bank Int'l v. Alice Corp. Pty. Ltd.*, 717 F.3d 1269 (Fed. Cir. 2013) (*en banc*).

Brief of Amici Curiae Law, Business, and Economics Scholars in Support of Respondents, *Alice Corp. v. CLS Bank Int'l*, 134 S. Ct. 2347 (2014) (No. 13-298).

Brief of Amicus Curiae Gibbons Institute of Law, Science and Technology in Support of Neither Party, *CLS Bank Int'l v. Alice Corp. Pty. Ltd.*, 717 F.3d 1269 (Fed. Cir. 2013) (*en banc*).

Brief for Google Inc., et al. as Amici Curiae Supporting Respondents, *Alice Corp. v. CLS Bank Int'l*, 134 S. Ct. 2347 (2014) (No. 13-298).

Brief for Microsoft Corp., et al., Amici Curiae in Support of Affirmance, *Alice Corp. v. CLS Bank Int'l*, 134 S. Ct. 2347 (2014) (No. 13-298).

WORKING PAPERS AND UNPUBLISHED MANUSCRIPTS

Ben Dugan, Estimating the Impact of *Alice v. CLS Bank* Based on a Statistical Analysis of Patent Office Subject Matter Rejections (Feb. 23, 2016) (unpublished manuscript), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2730803.

Anne Layne-Farrar, Defining Software Patents: A Research Field Guide (Feb. 15, 2006) (unpublished manuscript), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1818025.

REPORTS AND STUDIES

Exec. Office of the President, Patent Assertion and U.S. Innovation (2013).

Stuart Graham & Saurabh Vishnubhakat, Of Smart Phone Wars and Software Patents, 27 J. Econ. Persp. 67 (2013).

U.S. Gov't Accountability Office, GAO-13-465, Intellectual Property: Assessing Factors That Affect Patent Infringement Litigation Could Help Improve Patent Quality (2013).

PERIODICALS AND MAGAZINES

Timothy B. Lee, Analysis: Microsoft's Software Patent Flip–Flop, *Ars Technica* (Mar. 13, 2007).

Timothy B. Lee, Will the Supreme Court Save Us from Software Patents?, *Wash. Post* (Feb. 26, 2014).

Brian J. Love, No: Software Patents Don't Spur Innovation, but Impede It, *Wall St. J.*, May 12, 2013, at R2.

Robert Merges, Symposium: Go Ask Alice—What Can You Patent after *Alice v. CLS Bank*?, *SCOTUSblog* (June 20, 2014).

Randall Stross, Why Bill Gates Wants 3,000 New Patents, *N.Y. Times* (July 31, 2005).

INDUSTRY PUBLICATIONS AND WEBSITES

Gina Smith, Unsung Innovators: Marty Goetz, Holder of First Software Patent, *Computerworld* (Dec. 3, 2007).

Mark Summerfield, How the Fate of Software and Business Method Patents has Turned on USPTO Directors and the Courts, *Patentology* (Jan. 8, 2017).

ORGANIZATION REPORTS AND PUBLICATIONS

Proposed Amendments to Patent Eligible Subject Matter under 35 U.S.C. § 101, Intell. Prop. Owners Ass'n (Feb. 7, 2017).

Chris Barry et al., 2013 Patent Litigation Study: Big Cases Make Headlines, While Patent Cases Proliferate, PricewaterhouseCoopers (2013).

ACADEMIC CONFERENCE PAPERS AND PRESENTATIONS

Kay H. Brodersen, et al., Inferring Causal Impact Using Bayesian Structural Time-Series Models, 9 Annals of Applied Stats. 247 (2015).

ONLINE RESOURCES AND DATABASES

Patents, Microsoft, http://www.microsoft.com/en-us/legal/intellectualproperty/patents/default.aspx.

Patent Technology Monitoring Team Report, Patent Statistics for Electrical Computers, Digital Processing Systems, Information Security, and Error/Fault Handling, U.S. Pat. & Trademark Office.

PatentCAM, IP Checkups, https://www.ipcheckups.com/software-solutions/.

SUPPLEMENTAL MATERIALS

The Bluebook: A Uniform System of Citation (Columbia Law Review Ass'n et al. eds., 20th ed. 2015).

Judge Richard Posner, *Reflections on Judging* (2013).

University of Washington School of Law, Law P547 -- Legal Protection of Computer Software Syllabus (Spring 2025).